

Towards an Automated Assessment of Deviation Desirability

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Abstract. Conformance checking techniques allow process analysts to identify whether process executions deviate from a process model. However, these techniques are not able to determine the desirability of these deviations, i.e., the degree to which stakeholders want their recurrence in the future. Therefore, analysts have to manually examine deviations to assess their desirability. This manual examination is labor-intensive and requires a substantial amount of domain knowledge, resulting in risks of human errors. To address this problem, this paper proposes a three-step framework that enables process analysts to set up a pipeline for automating the desirability assessment of deviations. Our framework focuses on the impact of deviations on one dimension (e.g., performance or outcome) at a time, returning whether this impact is positive, negative, or neutral. The framework should be applied to different impact dimensions, allowing analysts to see multiple facets of a deviations’ impact. Consequently, they can identify potential trade-offs between dimensions and prioritize the mitigation of more severe deviations. We demonstrate our framework in three different scenarios, showcasing its broad applicability.

Keywords: Process Mining · Conformance Checking · Deviation Desirability

1 Introduction

Conformance checking involves comparing traces of actual process behavior, recorded in an event log, with the expected behavior described by a process model [6]. This comparison enables process analysts to detect deviations, i.e., traces that diverge from the model [13]. Deviations differ in their impact on the process, which can be positive, neutral, or negative [16], as well as in the perceived significance of this impact by process stakeholders, which can range from low to high [28]. These factors together determine the desirability of deviations, which we define as the extent to which process stakeholders do or do not wish to see their recurrence in the future.

While conformance checking techniques enable process analysts to effectively detect deviations, they do not evaluate the desirability of those deviations [8, 16]. As

a result, analysts are often left to assess desirability in an ad-hoc, unstructured, and manual way [22]. Such assessments are prone to human error and carry the risk of false conclusions, e.g., when a logging error is mistakenly classified as a severe negative deviation [32]. These false conclusions, in turn, may lead analysts to investigate root causes and develop corrective actions for irrelevant deviations, resulting in wasted time and effort without solving the actual problems and sometimes even creating new ones [4, 28, 31].

To address this limitation and make deviation assessments more accurate and less labor-intensive, we propose a framework that enables analysts to set up an automation pipeline for such assessments. The framework builds on the manual approach to deviation desirability assessments introduced by Grohs et al. [16], and consists of three steps. First, an analyst selects a conformance checking technique for deviation detection. Second, the analyst quantifies the impact dimension of interest and defines an impact function that links the deviation impact to this dimension. Third, the analyst specifies a desirability function that assigns a desirability category based on the deviation impact.

By following these steps, the analyst sets up a desirability assessment pipeline that runs automatically, detecting deviations and categorizing them by desirability without requiring further input during subsequent analysis iterations. To demonstrate the applicability of our framework across different settings, we showcase its use in three scenarios. Specifically, our proof of concept features desirability assessment in alignment-based, declarative, and multi-perspective declarative conformance checking.

In the remainder, we introduce preliminaries in [Section 2](#), describe the proposed framework in [Section 3](#), provide a proof of concept in [Section 4](#), outline related work in [Section 5](#), and conclude by discussing implications and limitations in [Section 6](#).

2 Preliminaries

This section provides the necessary preliminaries on event logs, conformance checking techniques, and Conditional Average Treatment Effects (CATE), which we use in [Section 3](#) to illustrate a possible instantiation of our framework. In addition, we elaborate on the manual desirability assessment framework that our work builds on.

Event Logs. An event log is a collection of traces, where each trace represents a single execution of a process. A trace, in turn, is a sequence of events. Each event corresponds to a specific activity within the process and is recorded with a timestamp. To distinguish them, every trace has a unique ID and can have more attributes that provide context.

Conformance Checking Techniques. Conformance checking techniques can be divided into imperative and declarative ones, depending on the formalism that is used for specifying the model to which an event log is compared.

Imperative Techniques. Imperative techniques are based on process models that explicitly capture allowed activity sequences [9]. Common modeling notations are Petri nets and Business Process Model and Notation (BPMN). Conformance checking with these models can be done with token-replay or, the current state-of-the-art, trace alignments [6]. A trace alignment is a step-wise comparison of the trace and a possible execution path of the model. It consists of a sequence of steps, each made up from a pair of a trace event and a model activity, or an empty symbol $\gg$, if an event or activity

is without a counterpart [6]. If the trace event (model activity) is without a counterpart, this step is called a log (model) move, indicating an inserted (skipped) activity. Otherwise, the step is called a synchronous move. A cost function assigns costs to each step. An alignment is optimal if no other possible alignment has a lower cost.

Declarative Techniques. Declarative techniques rely on process models that do not explicitly capture the allowed execution sequences but rather impose a set of constraints on the process [9]. Thus, every trace is implicitly allowed as long as it does not violate any constraint. Commonly used constraints are precedence, response, initialization, or succession constraints and can be defined in modeling languages such as *Declare* [9]. It is also possible to define constraints that refer to not only the control-flow perspective, i.e., the ordering of activities, but also to other perspectives, such as time or resources. This can be done in multi-perspective extensions such as *MP-Declare* [9]. Corresponding conformance checking techniques check if a trace fulfills the set of constraints.

Conditional Average Treatment Effect (CATE). In this paper, we use CATE as a function that estimates the impact of deviations on a specific dimension of the process, e.g., the process performance. The CATE is defined as the expected difference in a specific outcome between treatment and control groups, conditional on a set of observed covariates [20]. The treatment group is thereby assigned a treatment whereas the control group is not treated. For example, a treatment in a healthcare process can be activity *admission to ICU* and the outcome could be patient survival. Then, CATE would show the change of survival likelihood when patients are admitted to the ICU. Thus, CATE is an estimation of the causal effect of a treatment feature on an outcome feature.

Previous Desirability Assessment Approach. Recently, Grohs et al. [16] proposed an approach for the manual assessment of deviation desirability in the context of conformance checking. It consists of five different steps, where the first three aim to eliminate false alarms, i.e., deviations that are detected but do not constitute any problem. After that, the fourth step evaluates the impact of deviations on four *impact dimensions*: compliance, process outcome, performance, and standardization. There are inherent trade-offs between these dimensions [12], which should be considered during assessment. Each deviation impact can be positive, negative, or neutral. If a deviation is found to have negative or positive desirability, a potential reaction is evaluated in the fifth step.

3 Desirability Assessment Framework

This section presents our framework for assessing the deviation desirability. Our framework should enable process analysts to set up a pipeline for the automated impact evaluation of deviations (the fourth step in [16]), classifying them as positive, negative, or neutral. The framework is applicable to any impact dimension that can be quantified using process-specific knowledge [16]. For example, a healthcare outcome could be quantified binary (healed/not healed), while compliance might use a discrete scale from 0 to 5. The performance dimension, relating to factors like cost, time, or quality [16], can be analyzed using the Devil’s Quadrangle [12]. This means splitting performance into sub-dimensions that involve trade-offs. For instance, deviating to ship goods by plane has a positive impact on time but a negative impact on costs. Once instantiated for a specific use case, the framework runs automatically in subsequent analyses.

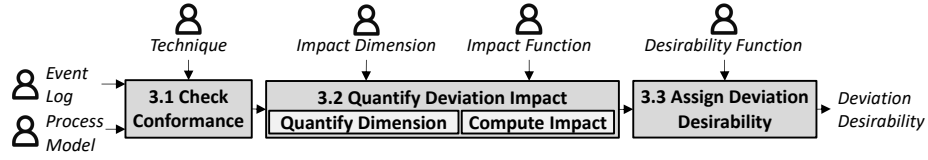


Fig. 1: Overview of our framework for evaluating the impact of conformance deviations

To set up the pipeline for deviation desirability assessment, our framework consists of three steps (cf. Fig. 1):

- (1) Checking the conformance between a process model and an event log (Sect. 3.1). For that, the conformance checking technique has to be specified.
- (2) Quantifying the impact of deviations by having the user select a dimension and define a custom impact function (Sect. 3.2).
- (3) Assigning deviation desirability categories (Sect. 3.3). This step requires a desirability function, which translates the deviation impact to desirability categories.

As a result, our framework outputs whether a deviation is positive, negative, or neutral regarding a chosen impact dimension. This should be repeated for all dimensions of interest, resulting in an assessment that shows possible trade-offs between dimensions.

In the following, we introduce the three framework steps in detail. We illustrate these using the invoice-to-cash example from Fig. 2 with process model and one trace.

3.1 Check Conformance

In the first step, a conformance check is conducted to identify which traces in the event log deviate from the process model, thus establishing the deviations to be analyzed. This can be done with different techniques, including trace alignments [6], declarative constraint checking [9], or high-level deviation pattern detection [15]. Our framework can also handle deviations from perspectives other than the control-flow, such as covered by checking for violations of *MP-Declare* constraints [9] on an event log. The selection of the concrete technique depends on the process complexity, the availability of corresponding process models and data in the event log, as well as the organizational context. For example, a very structured loan-application process can be modeled imperatively since it can be defined what exact behavior should occur whereas a complex healthcare process might be better modeled declaratively since the doctors should have flexibility.

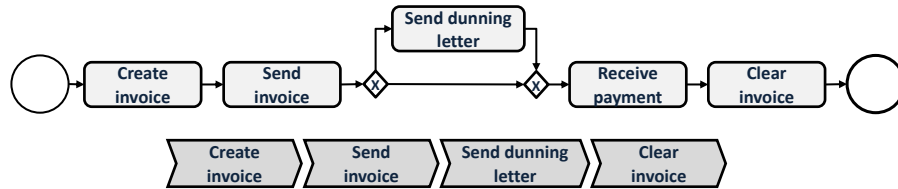


Fig. 2: Running example used to demonstrate our framework

The only assumption we make is that a deviation can be assigned to individual traces, which can be aggregated into a set of deviations that occur in an event log L , which we denote as $\mathcal{D}_L$. Based on that, we determine the existence of all detected deviations per trace. In particular, for each deviation $d \in \mathcal{D}_L$, each trace $t \in L$ is individually labeled depending on whether the deviation occurs in it (1) or not (0). For example, the trace in Fig. 2 skips the necessary *Receive Payment* activity, which will be detected as a deviation in the conformance check. When applying alignments, for instance, this leads to a model move on this activity and the trace is labeled with 1 for this model move. This results in an event log, in which each trace is associated with all deviations that occur in that trace.

3.2 Quantify Deviation Impact

In the second step, the impact of each deviation on the selected dimension is quantified. For that, the dimension is quantified and then the impact of each deviation on the dimension is computed.

Quantify Dimension. For each trace in the event log, we generate a numerical feature that represents the user-specified impact dimension. For instance, the time dimension can be quantified as the total trace duration. Other dimensions have to be quantified based on domain knowledge. As an example, the cost dimension can be quantified based on recorded attributes or other internal data sources, whereas the outcome dimension has to be defined based on the process goal, e.g., as a binary feature whether the trace outcome was positive (1) or negative (0). In our running example in Fig. 2, the outcome is not positive (quantified with 0), since no payment is received.

Compute Impact. After assigning each trace a value that quantifies the impact dimension of interest, our framework connects it to the previously detected deviations and quantifies the impact of each deviation on the chosen dimension. For that, it requires an *impact function* as input that takes the quantified dimension as well as the traces which are associated with all deviations and returns the impact for each deviation as a number. To illustrate our framework, we here use CATE as impact function, but in principle other methods like regression or decision trees could be used. Using CATE has the advantage that the quantified effects are causal rather than correlative. In our running example in Fig. 2, this CATE quantifies how the skip of *Receive Payment* affects the likelihood of a positive process outcome. Concretely, in our artificial example, this effect is -0.70, indicating that the deviation reduces the likelihood of positive outcomes by 70%.

In some cases, there might be other impact functions, e.g., if there is concrete information on how the deviation influences the impact dimension. For example, it might be contractually determined what monetary penalties occur if a certain deviation happens, thus increasing costs. In this case, the impact function can be defined as the sum over all cost due to the monetary penalties, the average of the incurred penalties over all traces, or a more complex function that involves several attributes to determine trace-based costs. Thus, the choice of the impact function can and should be adjusted to the specific use case. The result is a quantified impact that each deviation $d \in \mathcal{D}_L$ has on the impact dimension.

3.3 Assign Deviation Desirability

As the last step, a desirability category is assigned to each deviation based on the quantified impact from the previous step. To stay aligned with the output categories proposed in our previous work [16], we require the specification of a *desirability function*, which takes as input the impact value and outputs one of the desirability categories *neutral*, *positive*, and *negative*. One example would be a discrete function which assigns one of these categories to certain levels of impact, e.g., neutral is impact between 0.1 and -0.1, positive if ≥ 0.1 , and negative if ≤ -0.1 .⁴ In our running example in Fig. 2, the impact of skipping *Receive Payment* with -0.70 on the process outcome translates to a negative deviation, i.e., an undesirable one. The result is a desirability category for each $d \in \mathcal{D}_L$.

Users are recommended to apply our framework to multiple impact dimension as there are commonly trade-offs between them. For example, deviations might have positive impact on the time dimensions but negatively impact the outcome dimension. As an illustration, consider Tab.1 where deviations have been analyzed with a focus on outcome, time, and costs. Such a table can be the information upon which actions are taken by analysts. Users can, of course, also create desirability categories with custom names, e.g., calling high negative deviations critical. Such a multi-facet analysis can help to discover trade-offs between these dimensions as well as decide on how to proceed in mitigating undesired deviations. Once the analysis has been set up for all dimensions of interest, the assessment can be run automatically in subsequent analyses.

Deviation	Impact on	Outcome	Time	Costs	...
Model Move on Activity A		Low Negative	Medium Positive	Medium Negative	...
Log Move on Activity B		Low Positive	High Negative	High Positive	...
...		...	...	...	...

Table 1: Exemplary output of our framework w.r.t. to multiple impact dimensions

4 Proof of Concept

In this section, we demonstrate that our framework enables process analysts to set up a pipeline for automating the assessment of deviation desirability in different settings. To this end, we apply it to the real-world purchase order handling process from the BPI Challenge (BPIC) 2019, focusing on the *3-way match after Goods Receipt* variant of the process⁵.

⁴ Note that, in principle, any number of categories is possible as long as intervals of the impact function's range can be mapped to them. For example, a more detailed categorization could specify impact levels for positive and negative deviations as "low", "medium", and "high."

⁵ <https://doi.org/10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1>

Using this event log, we showcase that the framework applies to a diverse spectrum of conformance-deviations and impact dimensions:

1. Desirability of alignment-based deviations focusing on the time dimension.
2. Desirability of Declare violations focusing on the outcome dimension.
3. Desirability of MP-Declare constraint violations focusing on cost dimension.

These cover both conformance checking based on imperative and declarative models as well as control-flow-only and multi-perspective techniques⁶. The source code of our experiments can be found online.⁷

4.1 Impact of Alignment-Based Deviations on the Time Dimension

In the first setting, we investigated the impact of deviations based on trace alignments on the time dimension.

Check Conformance. Trace alignments require a process model as input, which is obtained from [11]. Computing the trace alignments between this process model and all traces in the event log with the corresponding PM4Py function [3] results in a set $\mathcal{D}_L$ that contains 47 log and model moves. This set includes, for example, a log move on activity *Change Price*, indicating that it was executed unnecessarily. Another example is the skip of activity *Record Goods Receipt*, as indicated by a model move on it.

Quantify Deviation Impact. To encode the time dimension, we utilize the total trace duration. This duration is computed for each trace in hours and ranges between 0 and 157,626 hours. Then, we calculate the CATE of each deviation $d \in \mathcal{D}_L$ on the trace duration to quantify the deviation impact. For that, we use the implementation of the *DoWhy* package [26], which allows for the estimation of causal effects, thereby accounting for potential effects that stem from other deviations. This CATE describes how much longer or shorter the duration of a trace is when it contains the deviation at hand compared to a trace that does not contain the deviation. This results in a set of 47 CATEs, one for each deviation. For example, the log move on *Change Price* increases the trace duration by 82.9 days, which indicates that this rework negatively impacts the time dimension. In contrast, the model move on *Record Goods Receipt* reduces the trace duration by 59.0 days. This indicates positive impact on the time dimension but likely also negatively impacts the outcome as a crucial activity in the process is skipped.

Assign Deviation Desirability. As the last step, we assign a desirability category to each of the deviations based on their CATE on the trace duration. For that, we define a discrete desirability function which assigns neutral desirability in case of a CATE of 0, low, medium, and high positive desirability for CATEs of > 0 h, > 100 h, and $> 1,000$ h, respectively, as well as the same negative desirability for CATEs of < 0 h, < -100 h, and $< -1,000$ h, respectively. This results in 21, 9, and 1 deviations with high, medium, and low positive desirability, respectively, as well as 11, 4, and 1 deviation with high, medium, and low negative desirability, respectively. Consequently, process analysts can prioritize those deviations with a high positive or negative desirability in mitigation efforts.

⁶ The framework is also applicable with other techniques such as multi-perspective alignments.

⁷ https://github.com/michaelgrohs/deviation_desirability_assessment

4.2 Impact of Declare Violations on the Outcome Dimension

In the second setting, we assess the impact of declarative constraint violations on the process outcome.

Check Conformance. There is no publicly available set of valid declarative constraints for the BPIC 2019. Therefore, we mine this set using the approach outlined by Di Ciccio and Montali [9, Algorithm 1], focusing only on precedence and response constraints.⁸ This results in 46 possible declarative constraint violations in $\mathcal{D}_L$, including, e.g., a precedence constraint that requires that activity *Vendor Creates Invoice* must have occurred before whenever *Remove Payment Block* happens.

Quantify Deviation Impact. Since the process outcome is not recorded in the BPIC 2019 logs, we define it based on the process. As the BPIC 2019 refers to purchase order handling, we consider a trace to have a positive outcome if the purchased goods are received (i.e., activity *Record Goods Receipt*) and the invoice is cleared (i.e., activity *Clear Invoice*), in that order. Otherwise, the trace is assigned a negative outcome.

After this, we again calculate the CATE of each deviation $d \in \mathcal{D}_L$ on the trace outcome to quantify the deviation impact. It describes the increased or decreased likelihood of a positive outcome for a trace when it contains the deviation at hand compared to a trace that does not contain the deviation. For example, if the precedence constraint between *Vendor Creates Invoice* and *Remove Payment Block* is violated, the likelihood of a positive outcome decreases by 2.04 %.

Assign Deviation Desirability. As for the previous setting, we use a discrete desirability function to assign a desirability category to each deviation. CATEs of > 0 , > 0.05 , and > 0.25 refer to low, medium, and high positive desirability (and equivalently for negative desirability). This results in 10 low and 21 medium negative deviations as well as 6, 8, and 1 low, medium, and high positive deviations, respectively. One interesting finding is that the violation of the precedence constraint between *Record Service Entry Sheet* and *Create Purchase Order Item* has a high positive desirability, increasing the likelihood of a positive outcome by 26.47 %.

4.3 Impact of MP-Declare Violations on the Cost Dimension

Last, we investigate the impact of MP-Declare constraint violations on costs.

Check Conformance. To obtain MP-Declare constraints, we enrich some of the previously mined constraints by time components. In particular, we assume that three constraints have to be fulfilled in a certain time frame:

1. *Create Purchase Order Item* has to occur within 90 days before *Clear Invoice*
2. *Vendor Creates Invoice* has to occur within 90 days before *Clear Invoice*
3. *Clear Invoice* has to follow *Record Invoice Receipt* within 90 days

Quantify Deviation Impact. Costs associated to deviations are not recorded in the BPIC 2019 log. Thus, we assume that costs will occur whenever the required time frame is exceeded. In particular, we assume that the organization has to pay a penalty of 5% of the purchase value for every seven days that exceed the time frame. If, for instance, a purchase order worth 1,000\$ records 97 days between *Record Invoice Receipt* and

⁸ Note that other constraint types are possible as well.

Clear Invoice, costs of 50\$ occur. In this manner, each of the traces are assigned costs that occur due to the deviation.

Based on the assigned costs, we quantify the impact of the three MP-Declare constraints with a simple impact function, which sums up all incurred costs per deviation. Unlike other use cases, such as the previous two settings, we do not require any more sophisticated impact functions since the costs are directly related to the deviation itself. More concretely, we do not have to approximate the causal effect of the deviation on the costs with CATE, but rather can compute it directly since the effect is known without statistical approximation. The result is the total sum of all costs that the organization has to pay in penalties for each deviation. For example, costs for Constraint 1 sum up to 8.6 Mio.\$.

Assign Deviation Desirability. Again, we define a discrete desirability function based on the total sum of costs incurred for a deviation. In particular, we consider costs below 100,000\$ to be low negative, between 100,000\$ and 1,000,000\$ medium negative, and anything above 1,000,000\$ high negative deviations. Positive deviations cannot occur as costs are negative by definition⁹. As a result, two of the three deviations are categorized as high negative and one as low negative, as illustrated in Tab. 2. A similar table can be constructed for all framework applications, including those from the previous two sections.

Deviation	Impact on	Costs	...
<i>Create Purchase Order Item</i> occurred more than 90 days after <i>Clear Invoice</i>		High Negative	...
<i>Vendor Creates Invoice</i> occurred more than 90 days after <i>Clear Invoice</i>		High Negative	...
<i>Clear Invoice</i> followed <i>Record Invoice Receipt</i> after more than 90 days		Low Negative	...

Table 2: Exemplary output of our framework for MP-Declare violations

5 Related Work

Our work relates to approaches that assess deviation desirability and to approaches that quantify effects in event data.

Assessing Deviation Desirability. Some approaches quantify deviation desirability using a numerical score. This is typically done by defining a cost function that measures the degree of non-conformance both within [1,27] and beyond the control-flow perspective [31]. Alternatively, desirability is expressed as a criticality ranking based on severity and likelihood of risks associated with the deviation [2]. While these approaches allow process analysts to rank detected deviations, they only measure desirability as a single number without accounting for potential trade-offs between different dimensions.

Other studies classify deviations using predefined desirability categories, such as exceptions versus anomalies [22] or true alarms versus false alarms [30]. These classifications are typically based on the importance of violated rules, where deviations from

⁹ Note that cost savings could be considered in alternative use cases, which could then also yield positive, i.e., desirable deviations

more important rules are considered more significant [4, 19], or on tolerance intervals defined by domain experts [32]. These qualitative approaches primarily distinguish between neutral and negative deviations, largely ignoring positive ones. Also, they do not recognize trade-offs between dimensions.

Our framework synthesizes the strengths of qualitative and quantitative approaches, enabling analysts to prioritize corrective actions effectively. From qualitative methods, it adopts the ability to classify deviations as positive, negative, or neutral. From quantitative methods, it incorporates the ability to rank deviations by impact, but uniquely does so within each impact dimension. This nuanced ranking allows analysts to identify trade-offs between different dimensions and prioritize mitigation efforts. For example, an analyst could first act to prevent critical negative deviations, then promote those with high positive impact, before addressing less severe negative issues.

Quantifying Effects in Event Data. In Section 4, we quantify the effects of deviations using a causal approach. Therefore, we also review two related streams of research.

Many approaches analyze process features through statistical correlation [23]. These works often aim to find reasons for deviations by correlating them with features like resources or geographical locations, frequently employing decision or regression trees [7, 10, 14, 18]. However, these correlation-based methods have two fundamental limitations. They do not quantify the impact of a deviation on dimensions like time or cost to assess its *desirability*, and more importantly, correlation does not imply causation.

To address the issue of causation, another research stream applies causal discovery techniques to process data. These use graph-based algorithms to uncover causal relationships [24, 25], probabilistic logic to find reasons for undesired behavior [29], statistical tests for treatment effects [21], or uplift trees for root cause analysis [5]. While powerful, these existing causal approaches have not been applied to the specific problem of evaluating deviation desirability, which is a gap our work aims to fill.

6 Discussion & Conclusion

In this paper, we propose a framework that enables process analysts to set up a pipeline for the automated assessment of the desirability of conformance deviations. It extends our previous work [16] by providing guidance on how to assess the desirability of true deviations, a step previously not covered in this detail. Especially the trade-offs between impact dimensions can be investigated with our framework, which is often not considered to this degree of granularity. Thus, a heatmap of deviation desirability as in Tab. 2 is a first step towards such a multi-facet analysis. The framework has to be instantiated and set up once, and can then be used automatically to update the results.

Limitations. Our work has several limitations. First, the framework itself is not automated, as it requires domain knowledge for the setup. Given the complexity of the task [16] and the need for flexibility across impact dimensions [17], users must tailor the aspects to their specific goals, context, and data. Our framework provides a structured pipeline to guide this process, but it could be extended by expert validations. Second, an impact dimension can be quantified in multiple ways (e.g., using different functions or data sources), potentially leading to contradictory results. We therefore do not propose a one-size-fits-all solution, as a single approach is unlikely to suit all orga-

nizational contexts and processes. Such contradicting results could be managed through domain-specific adaptations. Third, the impact function is intentionally not detailed in the framework as it needs to be tailored to specific dimensions and data. This limits our ability to account for confounding variables, a problem central to root cause analysis. It is important to note, though, that our work does not aim to extend that research stream, but rather aims to showcase that causal techniques can be effectively applied to the context of deviation desirability. This could be resolved by using more sophisticated causal discovery techniques. Fourth, our evaluation is a proof-of-concept rather than a full quantitative or qualitative study. This is because ground truth data is impossible to generate synthetically [16] and difficult to obtain from real organizations. We demonstrate the framework's applicability for practical use, rather than formally evaluating its usefulness to process analysts. In-depth user studies could help to address this limitation.

Future Work. In the future, we want to further investigate how to automate the assessment of deviation desirability. In particular, we want to also automate other steps from the manual framework proposed in [16] and, for instance, detect logging or model errors automatically. Further, we want to evaluate our framework in real-life settings by conducting a case study with experts who can validate the usefulness of our framework. We also want to assess whether the prioritization of more severe deviations is possible based on our framework, as well as whether mitigation efforts can be brought to life with it. One idea would be to define multi-objective recommendations automatically without manual definitions of actions. We also want to investigate an interactive way of defining impact and desirability functions based on recorded trace and event attributes, for which querying options in existing process mining platforms could be a valid starting point.

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